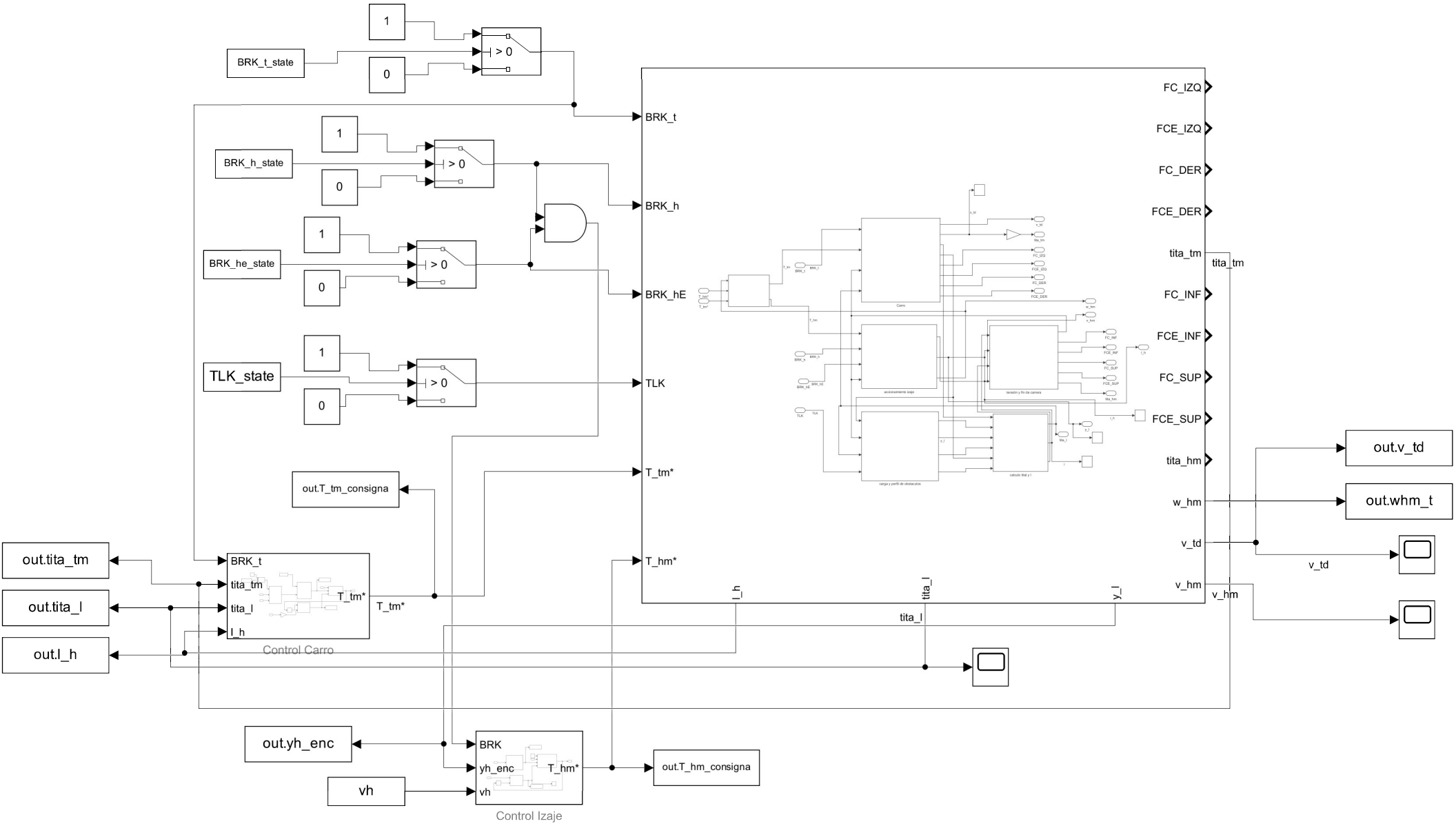
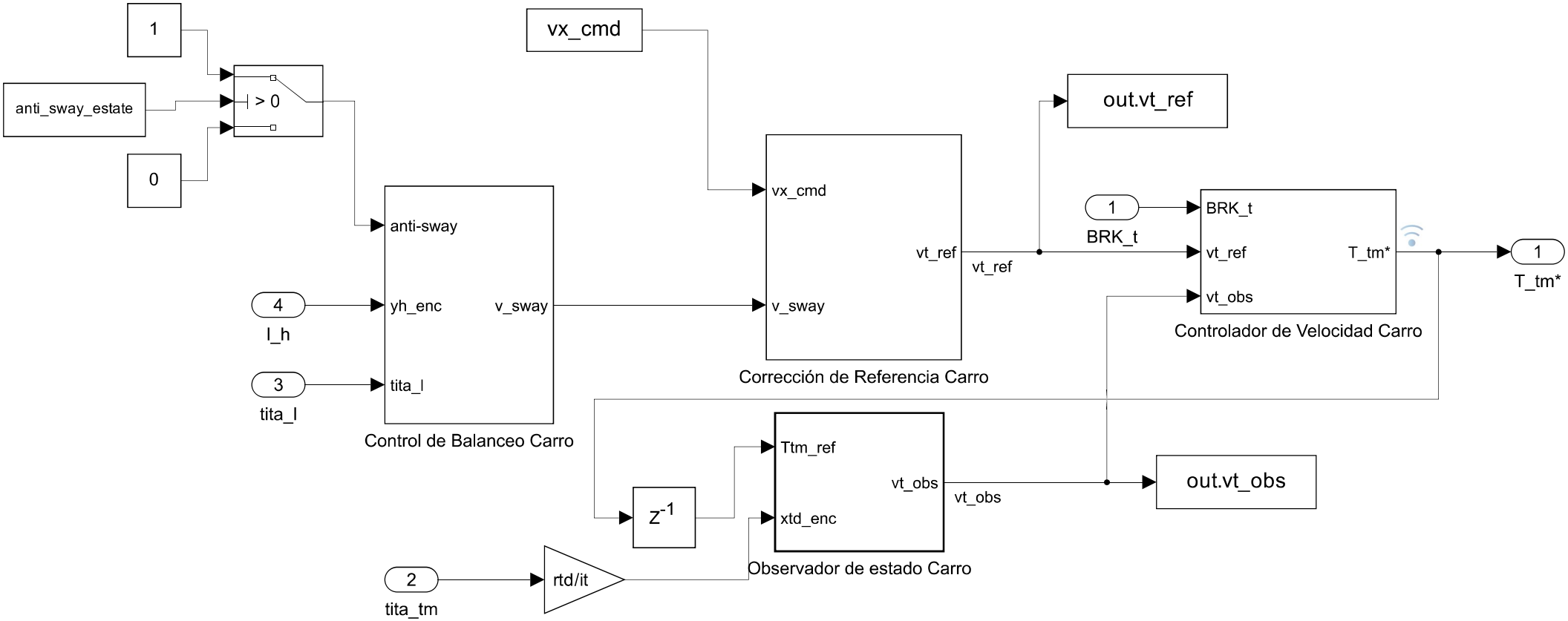
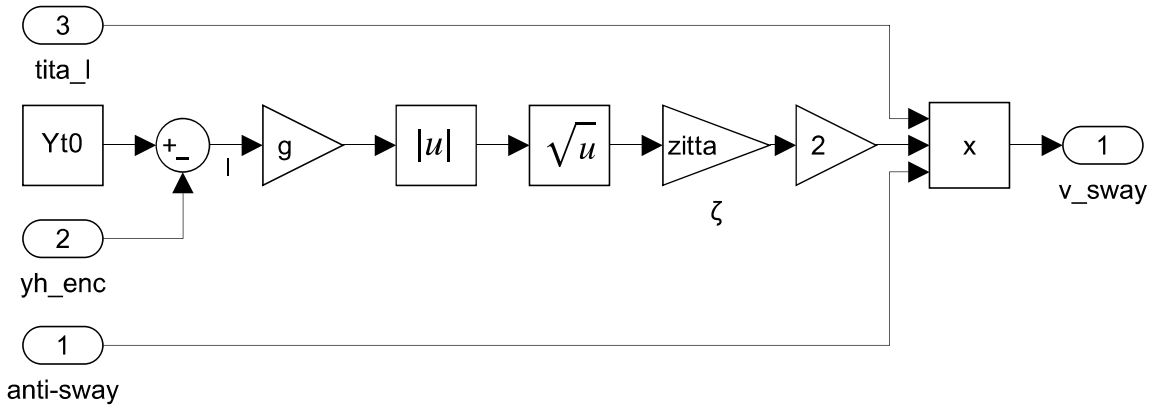
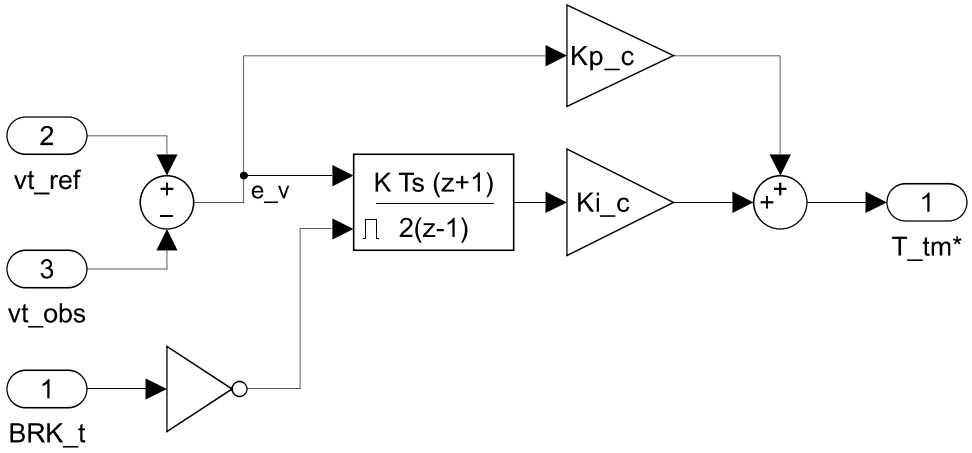
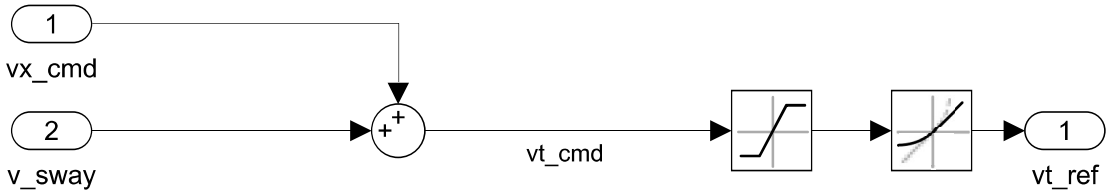
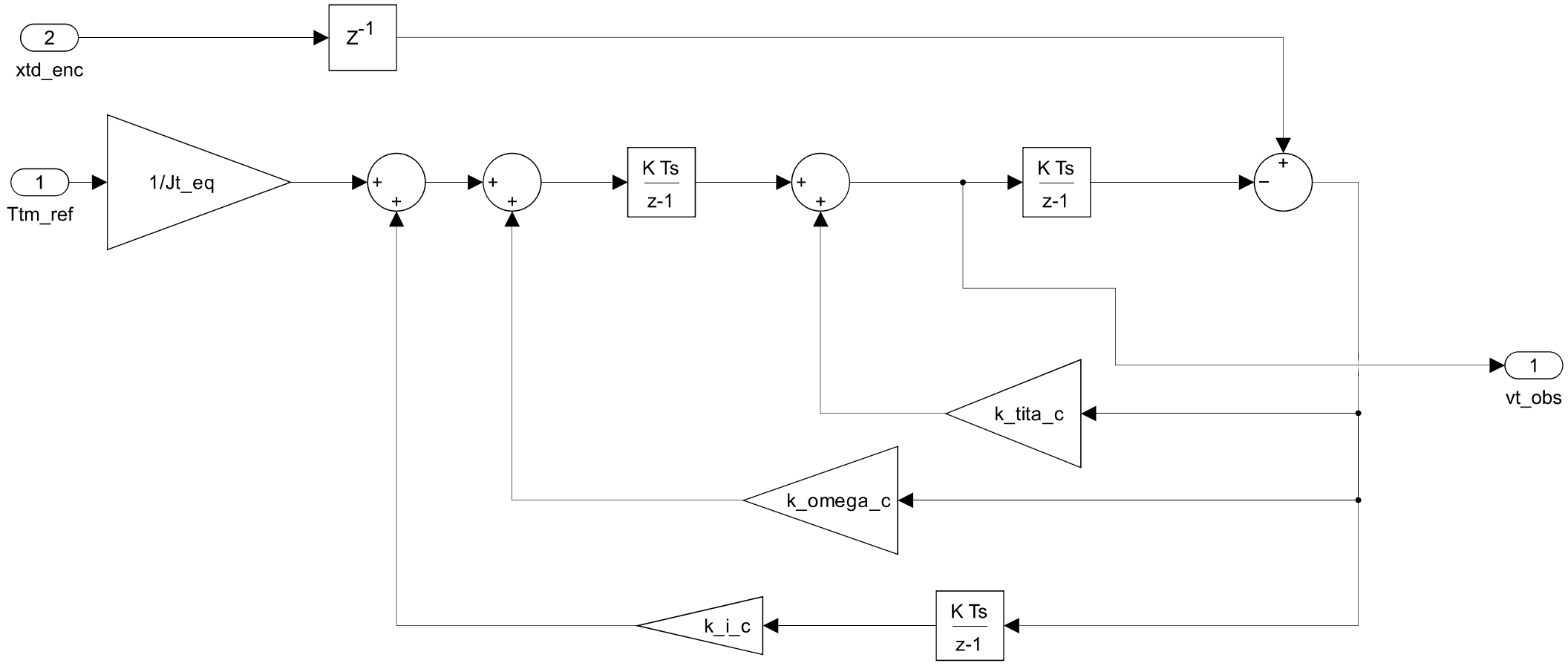

1
0
BRK_t_state
> 0
BRK_h_state
BRK_he_state
TLK_state
BRK_t
BRK_h
BRK_hE
TLK
T_tm*
T_hm*
out.T_tm_consigna
FC_IZQ
FCE_IZQ
FC_DER
FCE_DER
tita_tm
FC_INF
FCE_INF
FC_SUP
FCE_SUP
tita_hm
w_hm
v_td
v_hm
out.v_td
out.whm_t
out.tita_tm
out.tita_l
out.l_h
tita_l
Control Carro
out.yh_enc
vh
BRK
yh_enc
vh
Control Izaje
out.T_hm_consigna

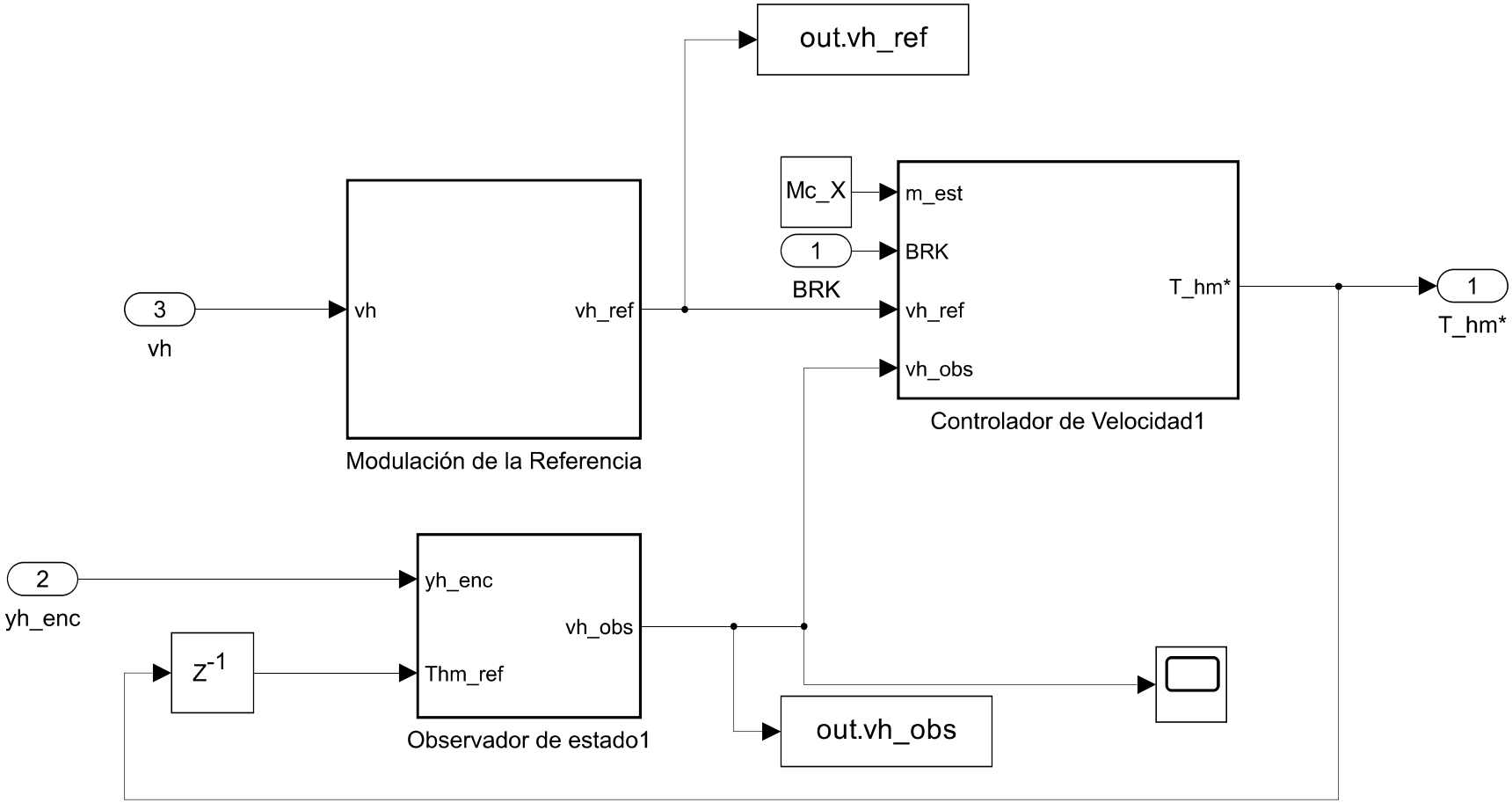

1
anti_sway_estate
0
> 0
vx_cmd
out.vt_ref
anti-sway
4
yh_enc
v_sway
l_h
3
tita_l
tita_l
Control de Balanceo Carro
vx_cmd
vt_ref
v_sway
vt_ref
Corrección de Referencia Carro
1
BRK_t
BRK_t
vt_ref
T_tm*
vt_obs
Controlador de Velocidad Carro
1
T_tm*
Ttm_ref
vt_obs
xtd_enc
Observador de estado Carro
vt_obs
out.vt_obs
Z-1
2
tita_tm
rtd/it

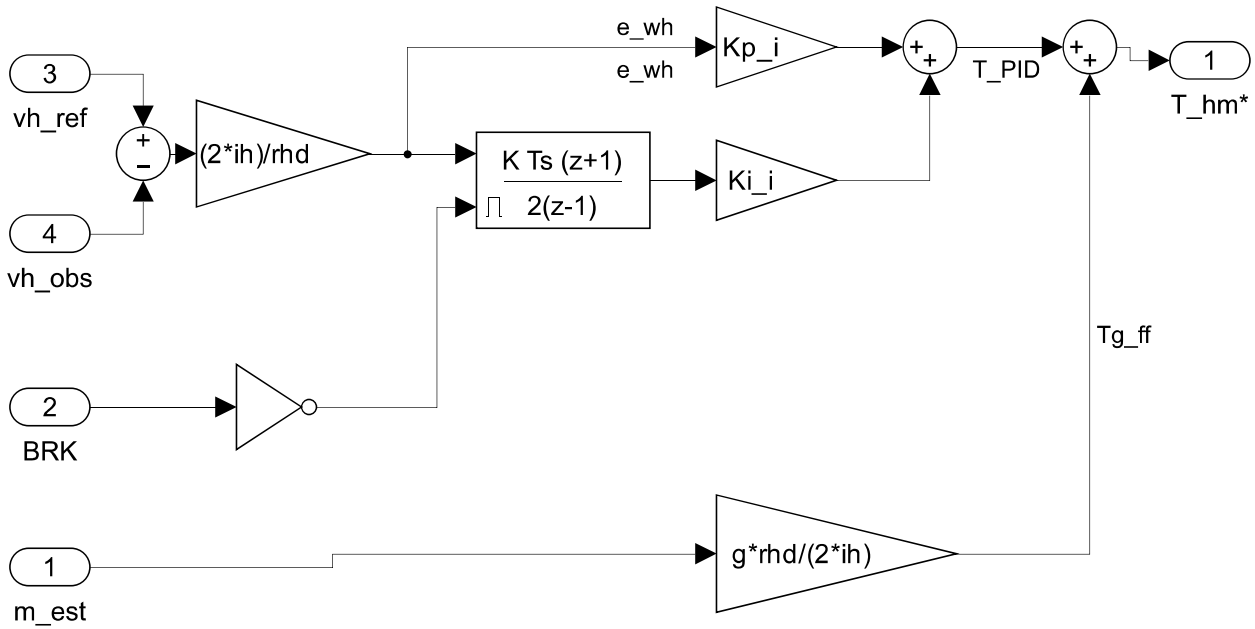

3
tita_l
Yt0
+_
l
g
|u|
√u
zitta
ζ
2
x
1
v_sway
2
yh_enc
1
anti-sway

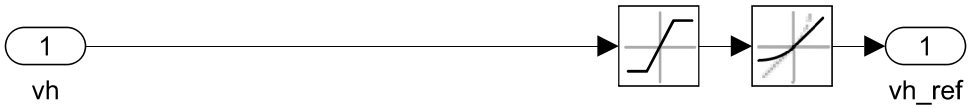

2
vt_ref
3
vt_obs
1
BRK_t
e_v
K Ts (z+1)
2(z-1)
Kp_c
Ki_c
1
T_tm*

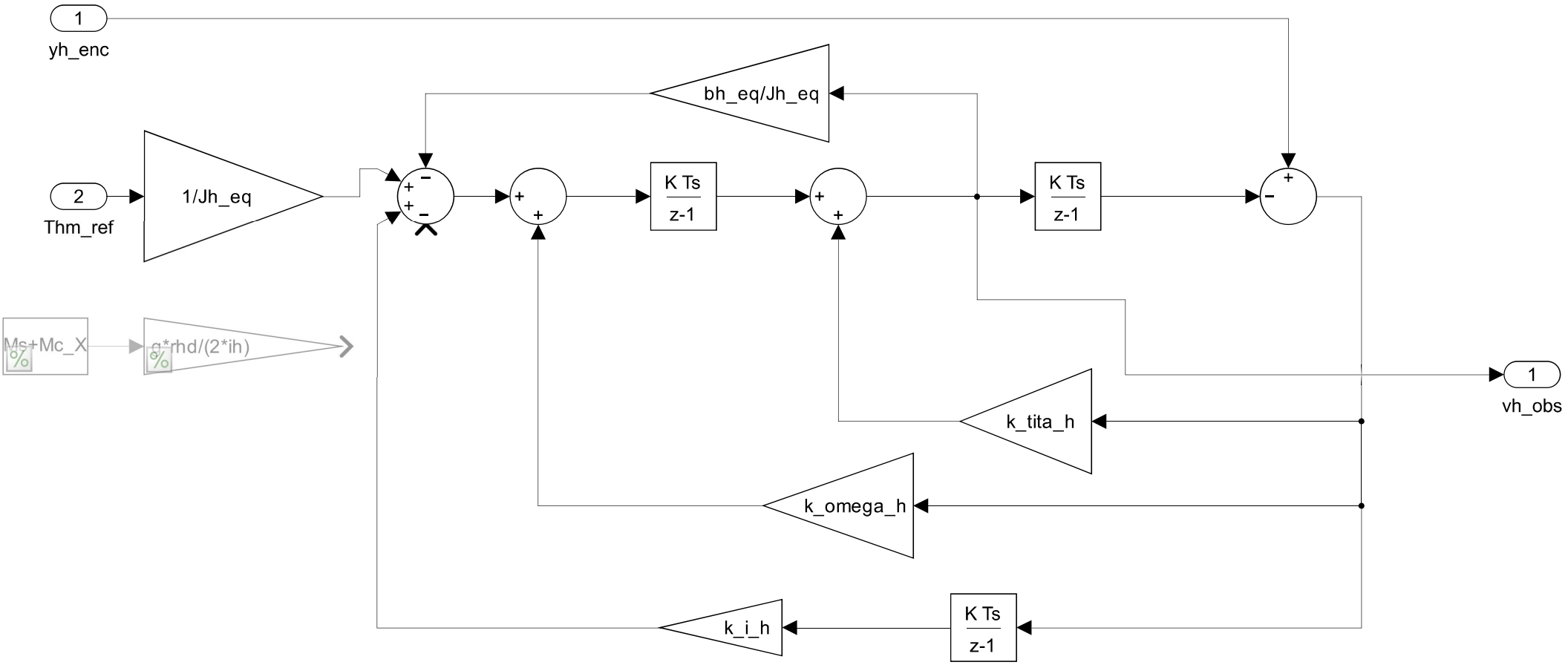

1
vx_cmd
2
v_sway
+
+
vt_cmd
1
vt_ref

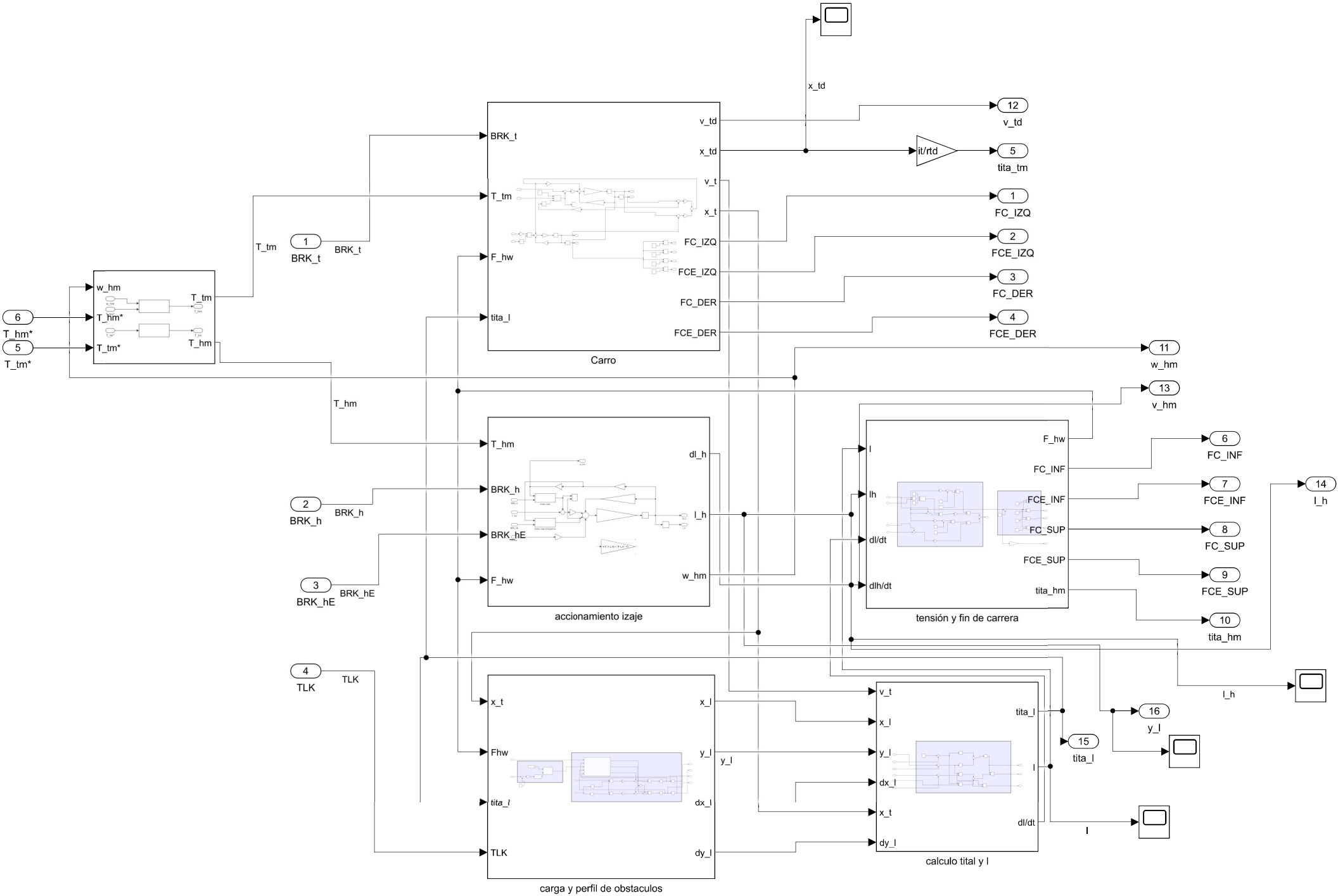

2
xtd_enc
Z^{-1}
1
Ttm_ref
1/Jt_eq
+
+
+
+
K Ts
z-1
+
+
K Ts
z-1
+
−
1
vt_obs
k_tita_c
k_omega_c
k_i_c
K Ts
z-1

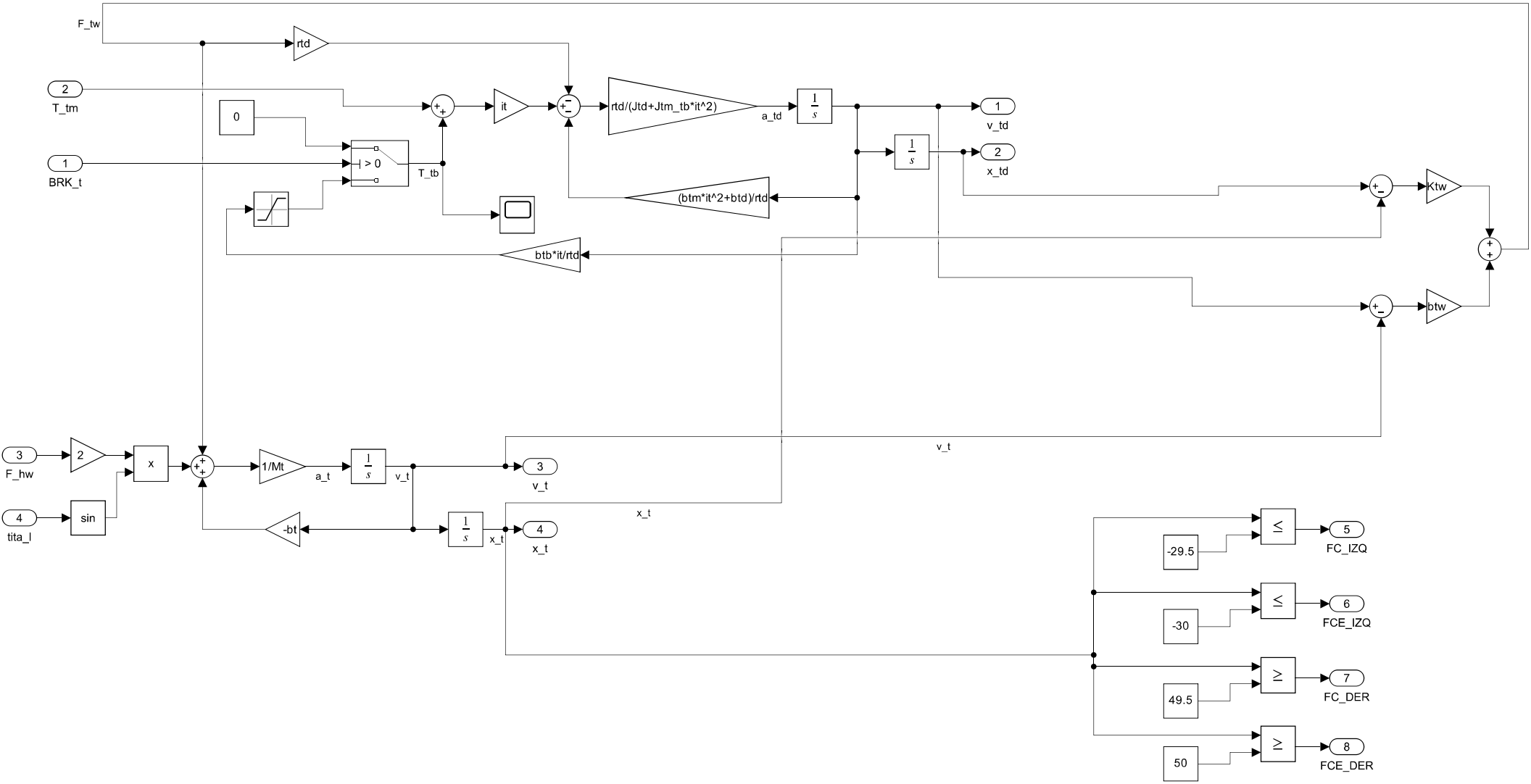

out.vh_ref
Mc_X
m_est
1
BRK
BRK
T_hm*
1
T_hm*
3
vh
vh
vh_ref
vh_ref
vh_obs
Controlador de Velocidad1
Modulación de la Referencia
2
yh_enc
yh_enc
vh_obs
z^{-1}
Thm_ref
Observador de estado1
out.vh_obs

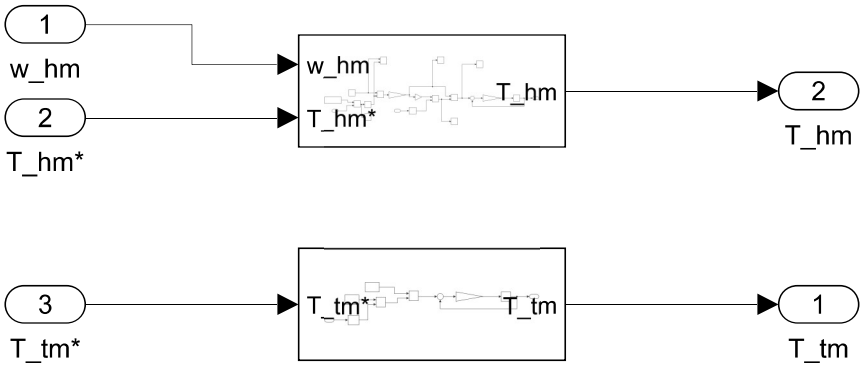

3
vh_ref
4
vh_obs
2
BRK
1
m_est
+
−
(2*ih)/rhd
e_wh
e_wh
Kp_i
K Ts (z+1)
2(z-1)
Ki_i
T_PID
1
T_hm*
Tg_ff
g*rhd/(2*ih)

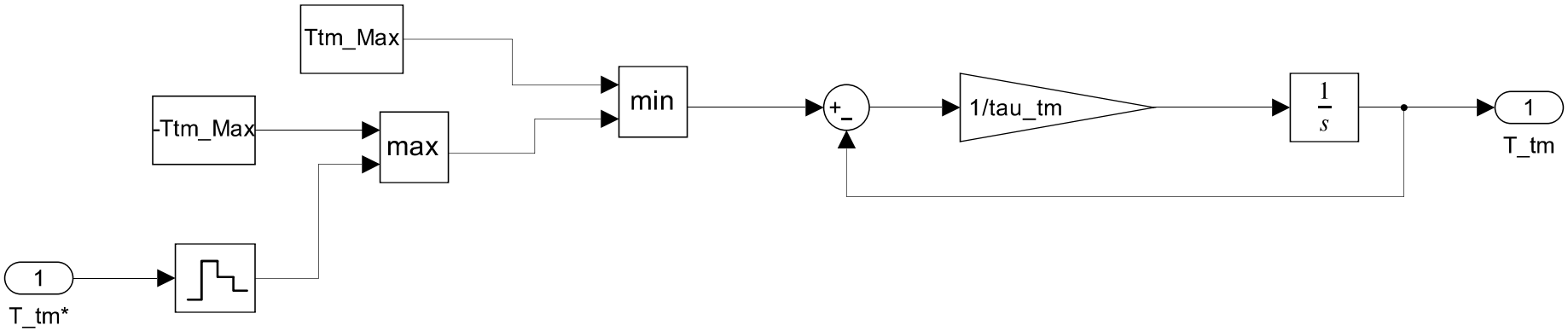

1
vh
1
vh_ref

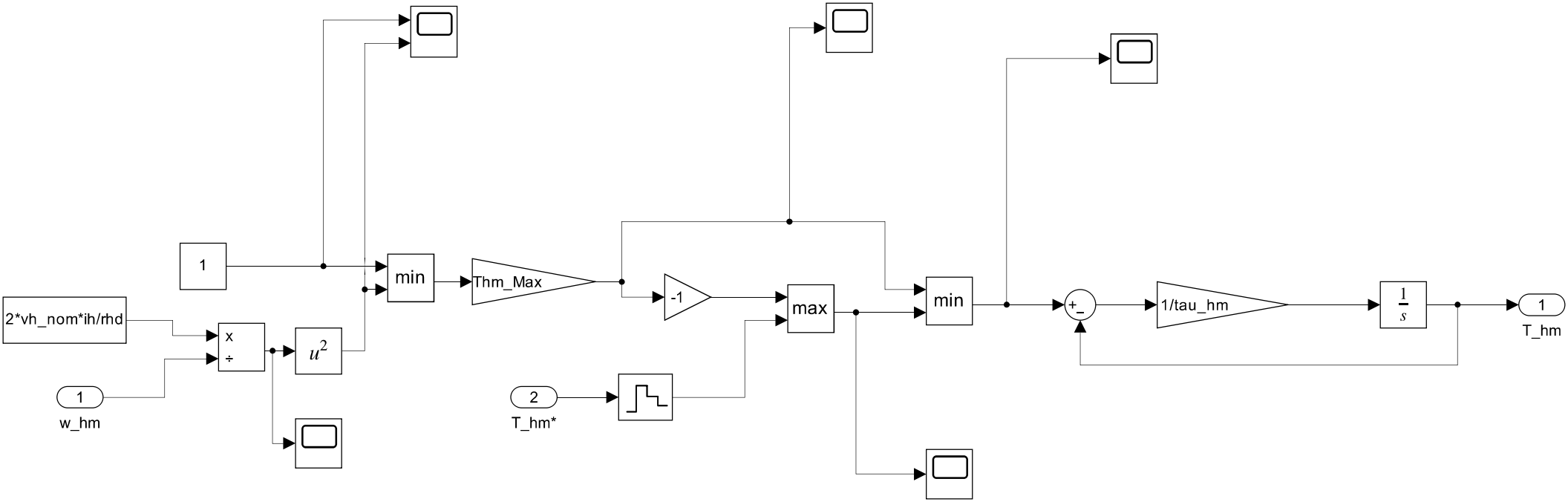

1
yh_enc
bh_eq/Jh_eq
2
Thm_ref
1/Jh_eq
K Ts
z-1
K Ts
z-1
Ms+Mc_X
g*rhd/(2*ih)
1
vh_obs
k_tita_h
k_omega_h
k_i_h
K Ts
z-1

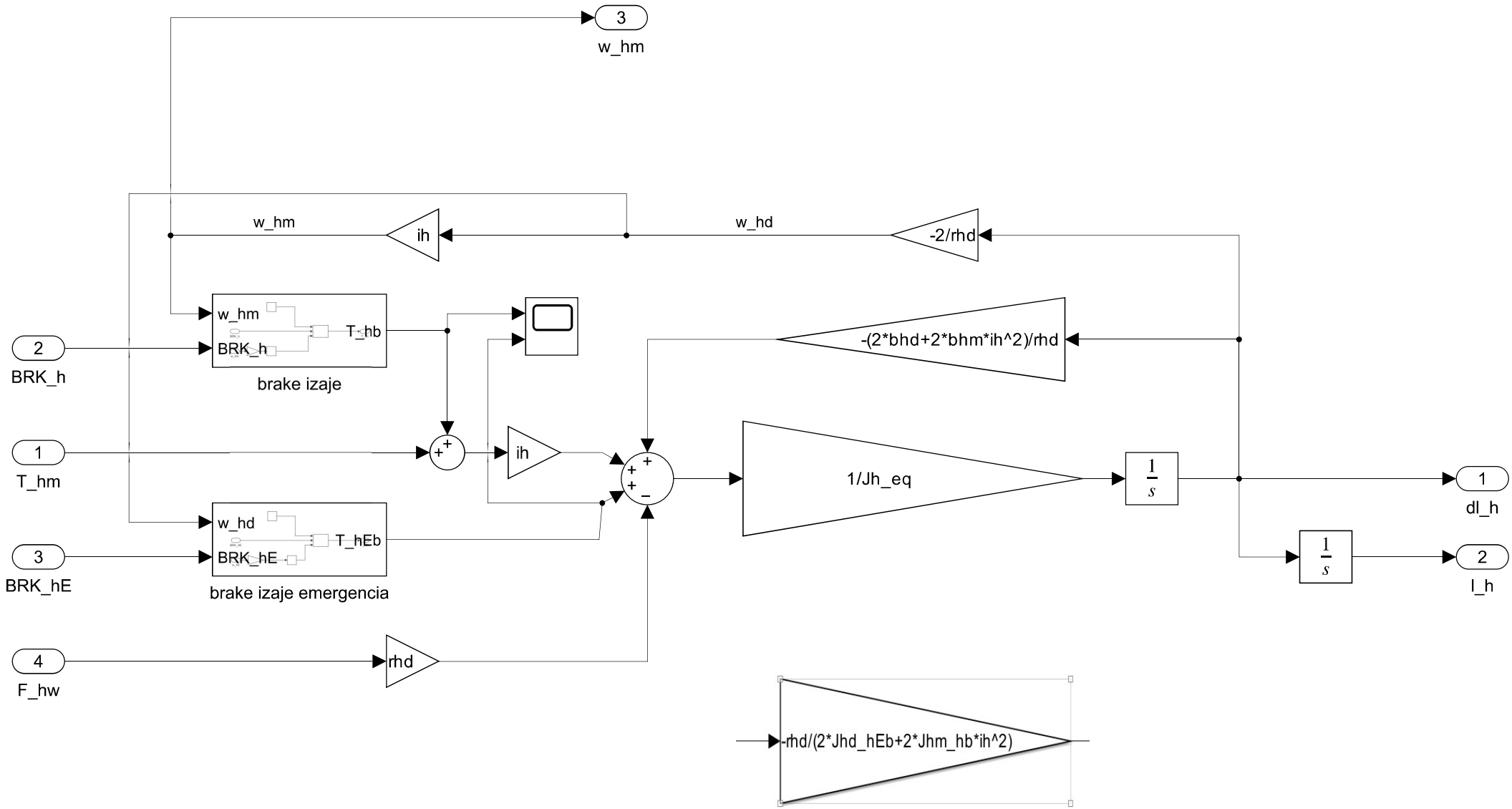

x_td
12
v_td
BRK_t
v_td
x_td
it/rtd
5
tita_tm
T_tm
v_t
1
FC_IZQ
1
BRK_t
BRK_t
x_t
2
FCE_IZQ
F_hw
FC_IZQ
3
FC_DER
FCE_IZQ
tita_l
4
FCE_DER
FC_DER
Carro
FCE_DER
w_hm
T_tm
6
T_hm*
T_hm*
5
T_tm*
T_hm
11
w_hm
13
v_hm
T_hm
T_hm
dl_h
6
FC_INF
BRK_h
BRK_h
2
BRK_h
l_h
7
FCE_INF
14
l_h
BRK_hE
FC_SUP
8
FC_SUP
3
BRK_hE
BRK_hE
F_hw
w_hm
FCE_SUP
9
FCE_SUP
accionamiento izaje
l
F_hw
FC_INF
tita_hm
10
tita_hm
lh
FCE_INF
tensión y fin de carrera
dl/dt
FC_SUP
FCE_SUP
dlh/dt
tita_hm
4
TLK
TLK
l_h
x_t
x_l
16
y_l
v_t
tita_l
Fhw
y_l
y_l
x_l
15
tita_l
tita_l
dx_l
y_l
l
x_t
l
dl/dt
dy_l
dx_l
TLK
dy_l
calculo tital y l
carga y perfil de obstaculos

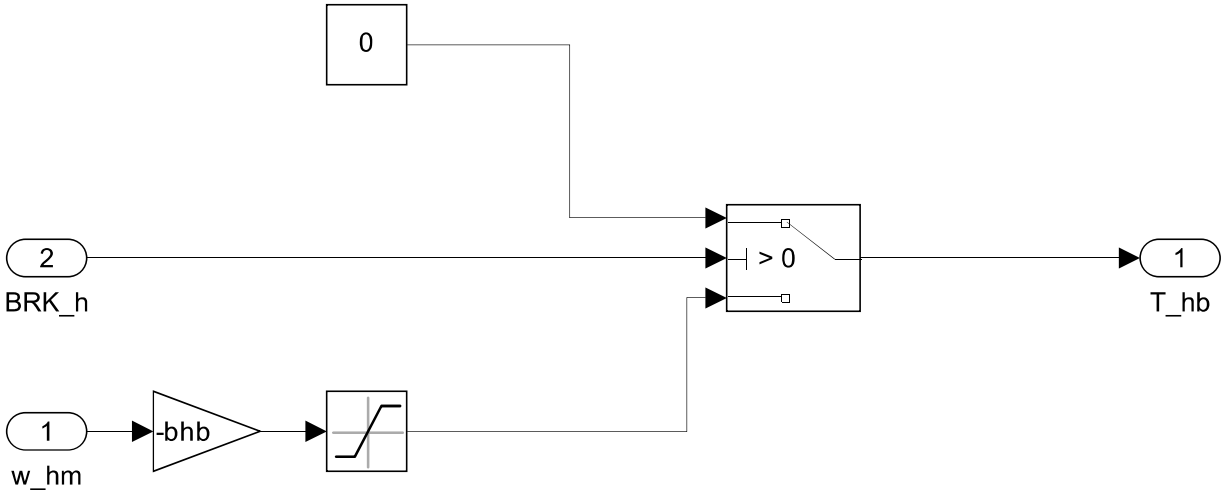

F_tw
rtd
2
T_tm
0
it
rtd/(Jtd+Jtm_tb*it^2)
a_td
1
v_td
1
BRK_t
> 0
T_tb
2
x_td
(btm*it^2+btd)/rtd
Ktw
btb*it/rtd
btw
3
F_hw
2
x
1/Mt
a_t
v_t
3
v_t
4
tita_l
sin
-bt
x_t
4
x_t
v_t
x_t
-29.5
5
FC_IZQ
-30
6
FCE_IZQ
49.5
7
FC_DER
50
8
FCE_DER

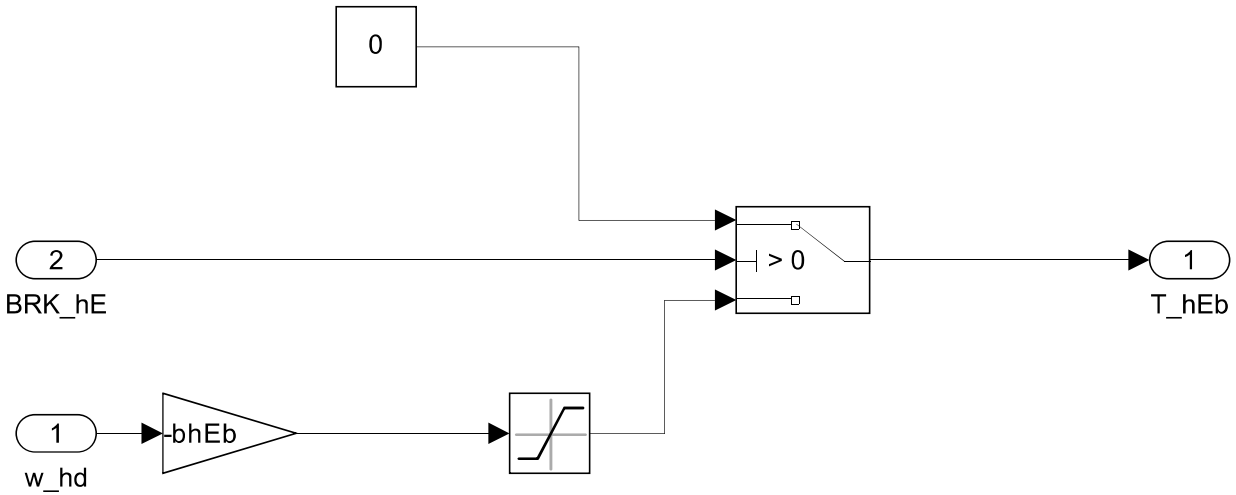

1
w_hm
2
T_hm*
w_hm
T_hm*
T_hm
2
T_hm
3
T_tm*
T_tm*
T_tm
1
T_tm

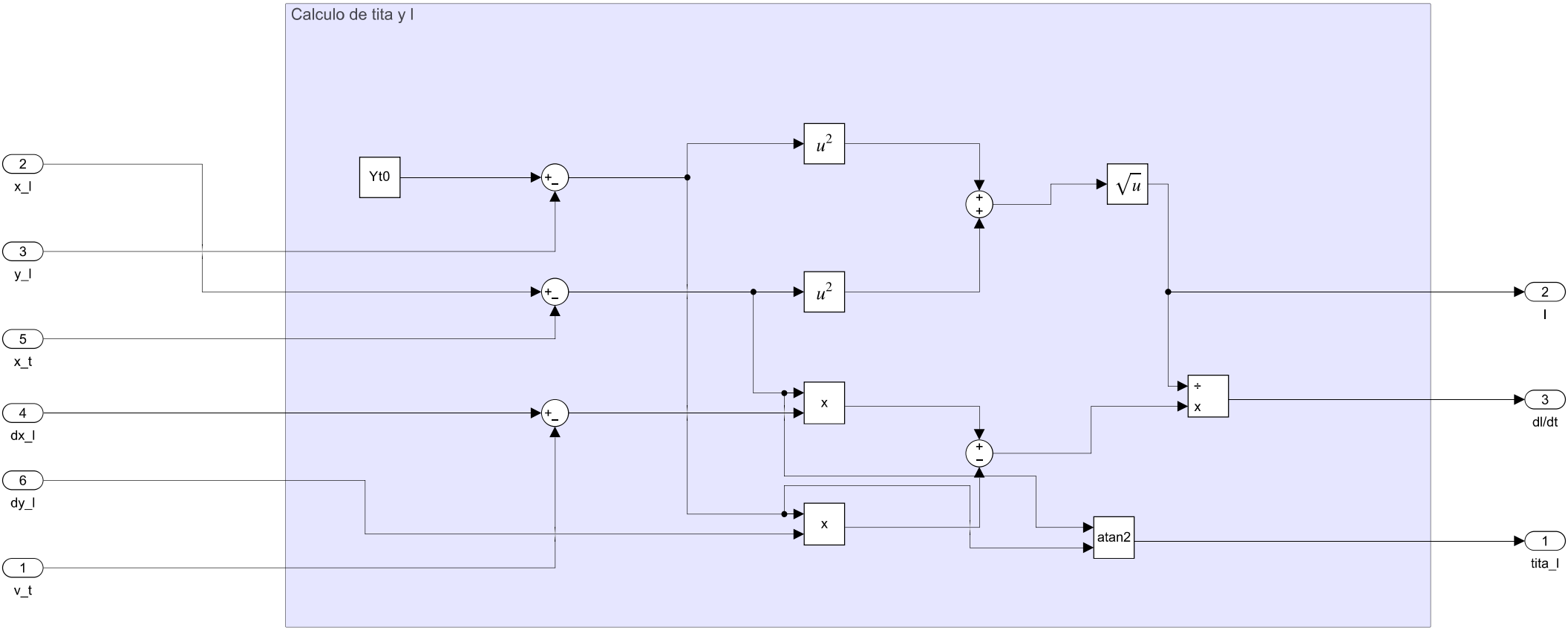

Ttm_Max
-Ttm_Max
max
min
+
_
1/tau_tm
1
s
1
T_tm
1
T_tm*

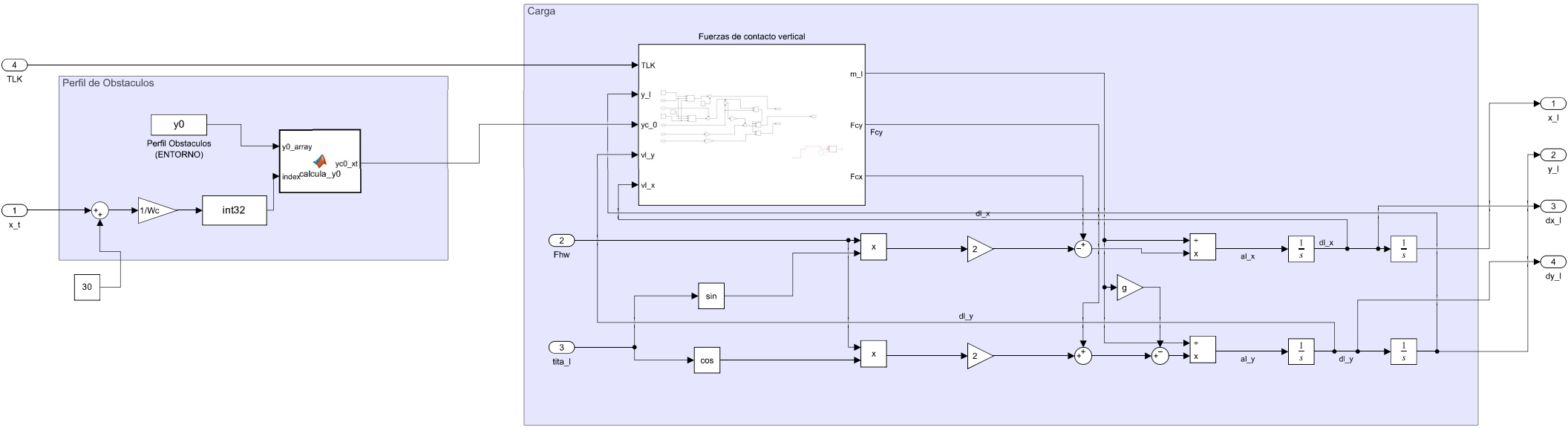

1
2*vh_nom*ih/rhd
x
÷
u^2
min
Thm_Max
-1
max
min
+_
1/tau_hm
$\frac{1}{s}$
1
T_hm
1
w_hm
2
T_hm*

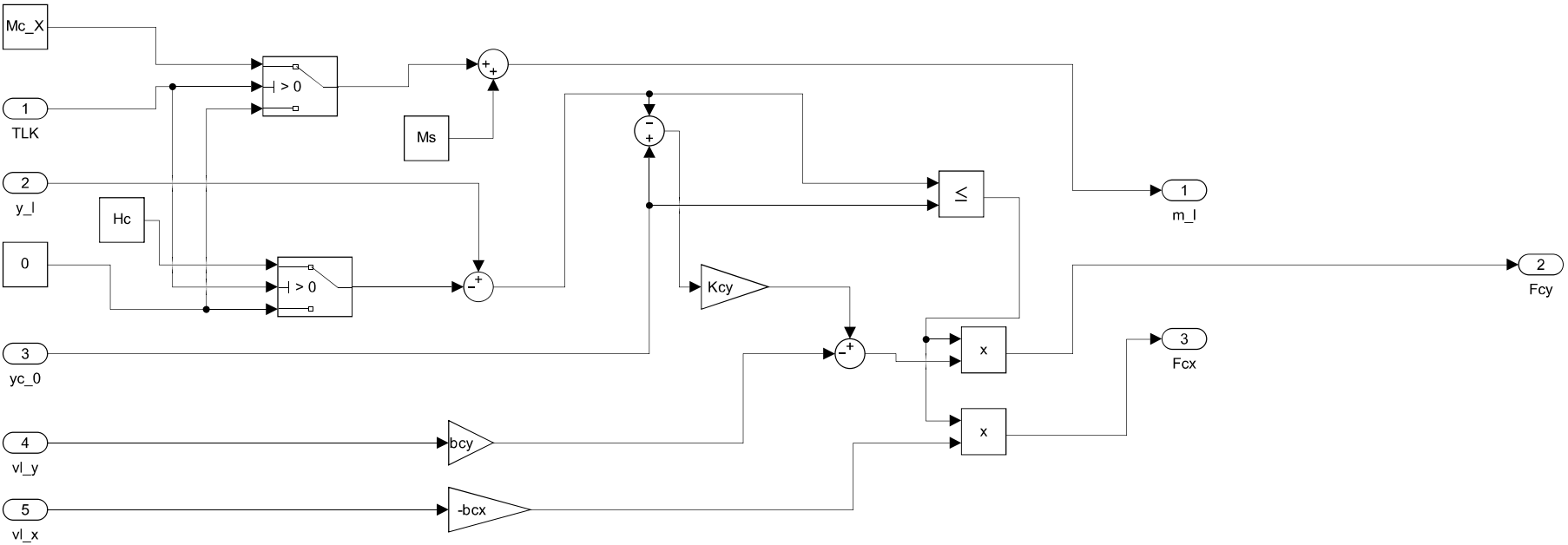

3
w_hm
w_hm
ih
w_hd
-2/rhd
w_hm
BRK_h
T_hb
brake izaje
2
BRK_h
-(2*bhd+2*bhm*ih^2)/rhd
1
T_hm
ih
1/Jh_eq
1
s
1
dl_h
w_hd
BRK_hE
T_hEb
3
BRK_hE
brake izaje emergencia
1
s
2
l_h
4
F_hw
rhd
-rhd/(2*Jhd_hEb+2*Jhm_hb*ih^2)


0
2
BRK_h
> 0
1
T_hb
1
w_hm
-bhb

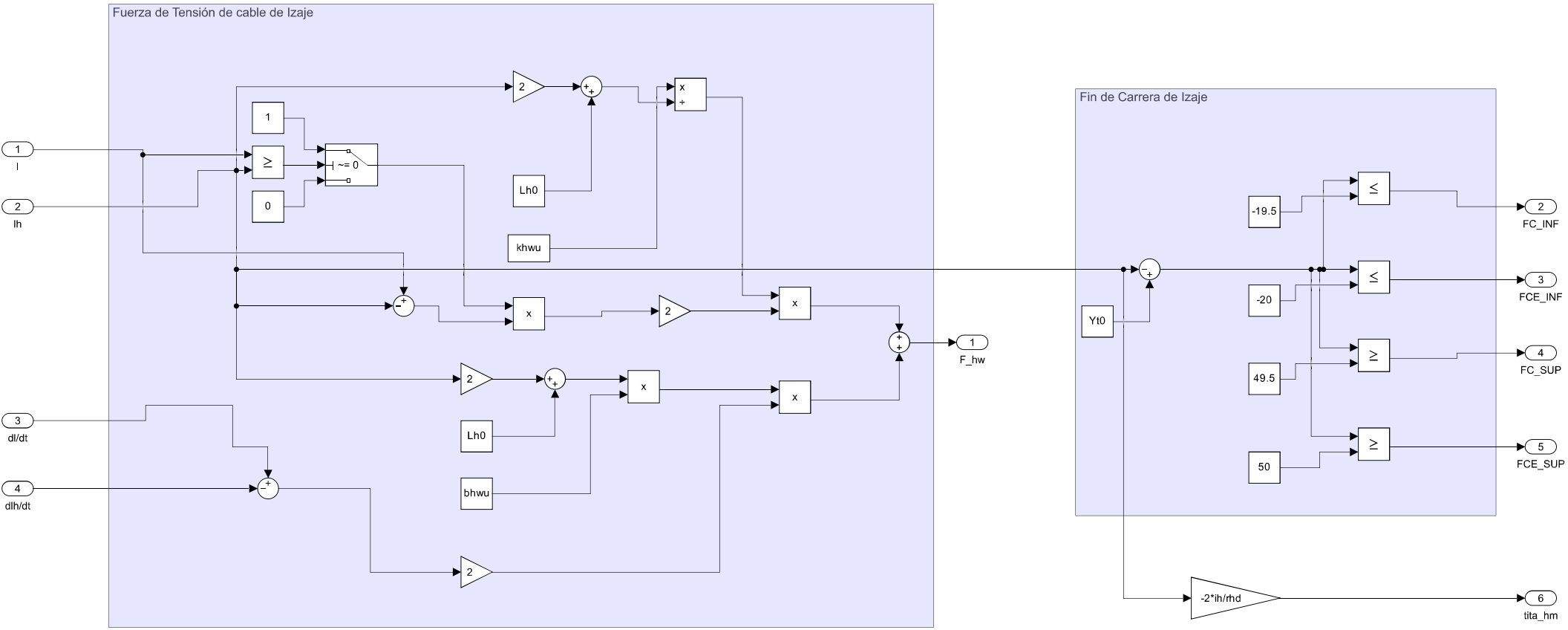

0
2
BRK_hE
> 0
1
T_hEb
1
w_hd
bhEb


Calculo de tita y l
2
x_l
3
y_l
5
x_t
4
dx_l
6
dy_l
1
v_t
Yt0
+_
+_
+_
u^2
u^2
x
x
++
+−
$\sqrt{u}$
÷
x
atan2
2
l
3
dl/dt
1
tita_l


4
TLK
Perfil de Obstaculos
y0
Perfil Obstaculos
(ENTORNO)
y0_array
yc0_xt
index
calcula_y0
1
x_t
1/Wc
int32
30
Carga
Fuerzas de contacto vertical
TLK
y_l
yc_0
vl_y
vl_x
m_l
Fcy
Fcx
dl_x
2
Fhw
x
2
÷
x
al_x
1/s
dl_x
1/s
g
sin
dl_y
3
tita_l
cos
x
2
+
−
÷
x
al_y
1/s
dl_y
1/s
1
x_l
2
y_l
3
dx_l
4
dy_l


Mc_X
1
TLK
2
y_l
Hc
0
3
yc_0
4
vl_y
5
vl_x
> 0
> 0
Ms
bcy
-bcx
Kcy
≤
x
x
1
m_l
2
Fcy
3
Fcx

```
function yc0_xt= calcula_y0(y0_array,index)
yc0_xt = y0_array(index);
```


Fuerza de Tensión de cable de Izaje
1
l
2
lh
3
dl/dt
4
dlh/dt
1
F_hw
Lh0
khwu
Lh0
bhwu
0
~= 0
Fin de Carrera de Izaje
Yt0
-19.5
-20
49.5
50
2
FC_INF
3
FCE_INF
4
FC_SUP
5
FCE_SUP
-2*ih/rhd
6
tita_hm

Page		System Name
1		controlador_torque
2		controlador_torque/Control Carro
3		controlador_torque/Control Carro/Control de Balanceo Carro
4		controlador_torque/Control Carro/Controlador de Velocidad Carro
5		controlador_torque/Control Carro/Corrección de Referencia Carro
6		controlador_torque/Control Carro/Observador de estado Carro
7		controlador_torque/Control Izaje
8		controlador_torque/Control Izaje/Controlador de Velocidad1
9		controlador_torque/Control Izaje/Modulación de la Referencia
10		controlador_torque/Control Izaje/Observador de estado1
11		controlador_torque/Subsystem
12		controlador_torque/Subsystem/Carro
13		controlador_torque/Subsystem/Subsystem2
14		controlador_torque/Subsystem/Subsystem2/Subsystem
15		controlador_torque/Subsystem/Subsystem2/Subsystem1
16		controlador_torque/Subsystem/accionamiento izaje
17		controlador_torque/Subsystem/accionamiento izaje/brake izaje
18		controlador_torque/Subsystem/accionamiento izaje/brake izaje emergencia
19		controlador_torque/Subsystem/calculo tital y l
20		controlador_torque/Subsystem/carga y perfil de obstaculos
21		controlador_torque/Subsystem/carga y perfil de obstaculos/Fuerzas de contacto vertical
22	(SF)	controlador_torque/Subsystem/carga y perfil de obstaculos/MATLAB Function
23		controlador_torque/Subsystem/tensión y fin de carrera